MAIS PAC

IPhO 02.04: Celestial Mechanics and Gravitation · Lecture + Problems

Celestial Mechanics and Gravitation

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Newton's law of gravitation, and why the inverse square is special

Two point masses attract with

$$F = (Gm_1m_2)/(r^2), G = 6.674×10^(-11) {N,m^2,kg^(-2)}. \quad (1)$$

For a spherically symmetric body, the field outside is exactly that of a point mass at its centre (Newton's shell theorem), which is why planets can be treated as points. Inside a uniform sphere the field falls linearly, g(r) = g_0,r/R — the reason a tunnel through the Earth gives simple harmonic motion.

The potential energy of a pair is

$$U = -(Gm_1m_2)/(r), \quad (2)$$

with the zero at infinite separation. The minus sign is not decoration: it is what allows bound states, and it makes the total energy negative for every closed orbit.

2. Kepler's three laws

1. **Orbits are ellipses with the Sun at one focus.** (For a two-body system, both bodies orbit their common centre of mass.)
2. **Equal areas in equal times.** Equivalently, angular momentum about the central body is conserved.
3. **$T^2 \propto a^3$.** For the Solar System in SI units,

$$T^2 = (4\pi^2)/(GM)a^3. \tag{3}$$

The third law is the fastest way to weigh a central body: measure a and T for anything orbiting it and M follows immediately.

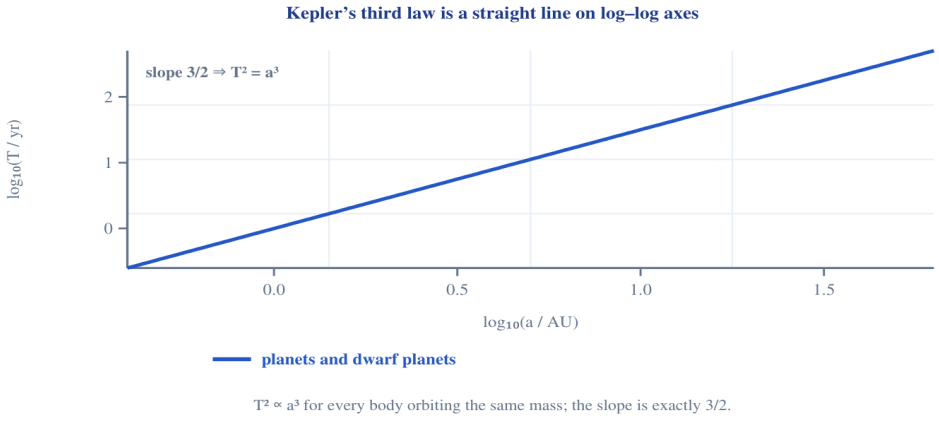


Figure. $T^2 \propto a^3$ for every body orbiting the same mass; the slope is exactly 3/2.

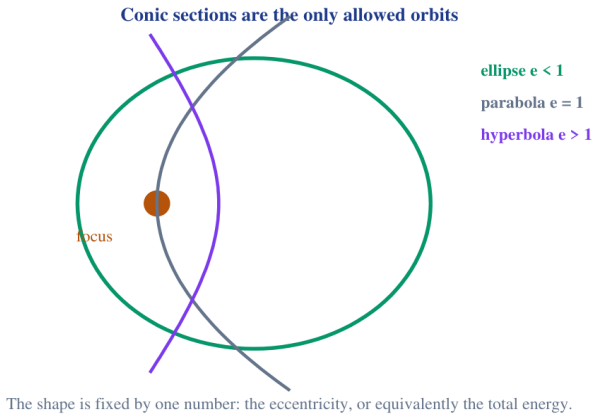


Figure 1. Bound orbits close (E < 0), parabolic orbits just escape (E = 0), and hyperbolic orbits always escape (E > 0).

3. The vis-viva equation: one line for every orbit

Energy conservation for a body of mass m in the field of mass M≫ m gives

$$\frac{1}{2} mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}, \quad (4)$$

hence the **vis-viva equation**

$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right). \quad (5)$$

From it everything follows:

- Circular orbit (r = a): $v_c = \sqrt{GM/r}$.
- Escape ($a \to \infty$, E = 0): $v_e = \sqrt{2GM/r} = \sqrt{2}\, v_c$.
- Ellipse: the speed is largest at periapsis and smallest at apoapsis.
- The period follows from the mean motion: $T = 2\pi\sqrt{a^3/GM}$.

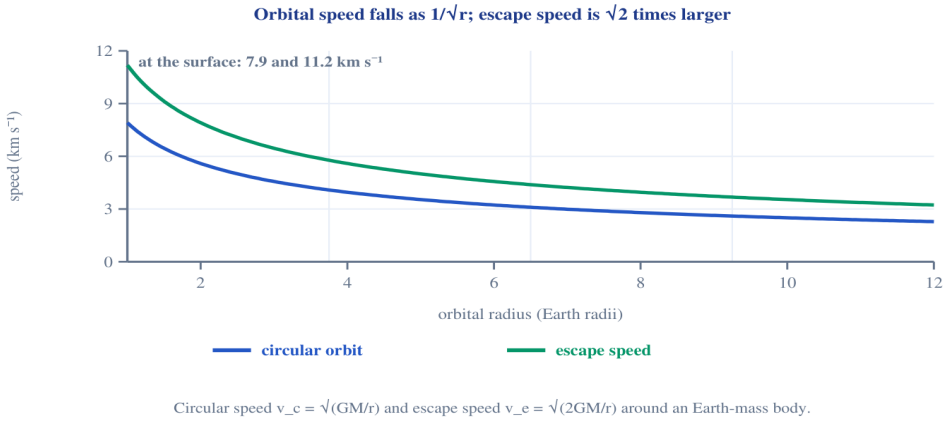


Figure. Circular speed $v_c = \sqrt{GM/r}$ and escape speed $v_e = \sqrt{2GM/r}$ around an Earth-mass body.

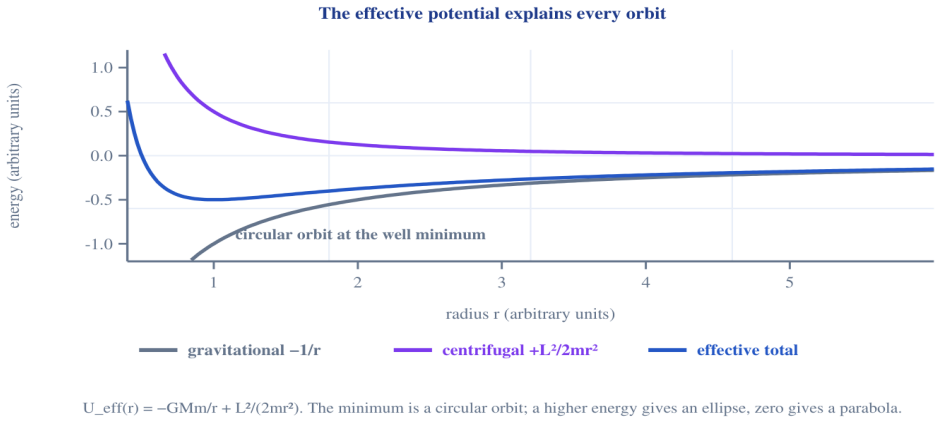
Interactive experiment: Orbit geometry: a, e and the vis-viva speed — open the online handout to use it.

4. The effective potential, and why orbits are stable

Angular momentum conservation reduces the two-body problem to one dimension. Writing $L = m r^2\dot{\theta}$,

$$E = \frac{1}{2} m \dot{r}^2 + \left\{\frac{L^2}{2mr^2} - \frac{GMm}{r}\right\}_{U_{\text{eff}}(r)}. \quad (6)$$

The effective potential has a single minimum, at $r_0 = L^2/(GMm^2)$. That minimum **is** the circular orbit. Small radial disturbances about it oscillate — which is why orbits are stable. The same formula explains why the angular-diameter distance and orbit periods behave the way they do in astronomy problems.



U_eff(r) = −GMm/r + L²/(2mr²). The minimum is a circular orbit; a higher energy gives an ellipse, zero gives a parabola.

Figure. U_eff(r) = –GMm/r + L²/(2mr²). The minimum is a circular orbit; a higher energy gives an ellipse, zero gives a parabola.

5. Two bodies, the barycentre, and Lagrange points

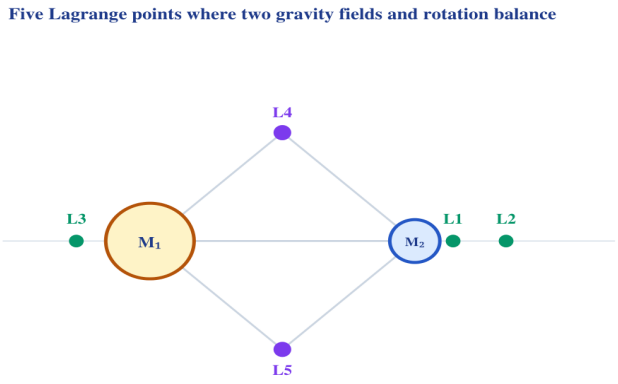
For two masses m_1 and m_2 separated by r, both orbit the barycentre, which lies at distance

$$r_1 = (m_2)/(m_1+m_2)r \tag{7}$$

from m_1. The relative motion obeys the same Kepler problem with M = m_1+m_2 — this is why a binary star's period gives the **total** mass, while the two velocity amplitudes give the ratio.

In a rotating two-body system there are five points where a small body can remain at rest relative to the pair:

- **L1, L2, L3** lie on the line joining the masses; they are unstable saddles.
- **L4, L5** form equilateral triangles with the two masses; they are stable when m_1/m_2 25, which is why Jupiter's Trojan asteroids collect there.



L1–L3 are unstable saddles; L4 and L5 are stable when the mass ratio exceeds about 25.

Figure 5. A small body at a Lagrange point corotates with the two larger masses; L4 and L5 can trap dust, asteroids and spacecraft.

The gravitational problem-solving kit. For any orbit question: (1) decide whether the speed, the energy or the period is asked; (2) write vis-viva or energy conservation; (3) use angular momentum for the “equal areas” statements; (4) use $T^2 = 4\pi^2 a^3/GM$ whenever a mass is wanted; (5) check the limit e→0 (circle) and E→0 (escape).

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Weighing a planet from its moon. A moon orbits a planet in a circular orbit of radius 1.2×10^8 {m} with period 4.0×10^5 {s}. (a) Find the planet's mass. (b) Find the moon's orbital speed. (c) Find the planet's density if its radius is 5.0×10^7 {m}, and comment.

Worked solution →

[1] Problem 2 — Escape speed and the atmosphere.

(a) Derive $v_e = \sqrt{2GM/R}$ from energy conservation. (b) Compute it for Earth and for the Moon ($M = 7.35\times10^{22}$ {kg}, $R = 1.74\times10^6$ {m}). (c) Explain why the Moon has no significant atmosphere while Earth retains nitrogen and oxygen, using rms molecular speeds at 300 K.

Worked solution →

[2] Problem 3 — Ellipse to circle to escape. A satellite orbits Earth circularly at radius r_0. A tangential burn changes its speed instantaneously by a factor α. (a) Show that the new semi-major axis is $a = r_0/(2-\alpha^2)$. (b) Find α for a parabolic escape. (c) Find the periapsis and apoapsis for $\alpha = 1.2$, and explain what happens for $\alpha > \sqrt{2}$ and for α slightly less than 1.

Worked solution →

[2] Problem 4 — Kepler's second law, quantitatively. For an elliptical orbit with semi-major axis a and eccentricity e, (a) write the angular momentum per unit mass in terms of a, e and M; (b) show that the areal velocity is constant and equals L/(2m); (c) prove that the speed at periapsis and apoapsis satisfy $v_p/v_a = (1+e)/(1-e)$.

Worked solution →

[2] Problem 5 — Geostationary orbit and launch cost.

(a) Find the radius of the orbit in which a satellite has a period of exactly one sidereal day (86164 {s}). (b) Find its height above the equator and its speed. (c) Find the minimum launch energy per kilogram from Earth's surface to that orbit, and compare with the kinetic energy per kilogram in low Earth orbit.

Worked solution →

[2] Problem 6 — Tunnelling through the Earth. A uniform-density Earth has radius R and mass M. A straight tunnel is bored between two surface points. (a) Show that the gravitational field at radius r inside is g = g_0 r/R. (b) Show that the force on a mass in the tunnel is proportional to its distance from the midpoint, and find the period of oscillation. (c) Evaluate the period numerically and explain why it does not depend on the tunnel's direction.

Worked solution →

[2] Problem 7 — Binaries and the barycentre. Two stars of masses M and 2M are 9.0×10^(10) {m} apart and orbit their common centre of mass. (a) Find the distance of each star from the barycentre. (b) Find the period. (c) Find each star's orbital speed, and explain why a single-lined spectroscopic binary cannot give the total mass without knowing the mass ratio.

Worked solution →

[2] Problem 8 — The Roche limit. A small satellite of density ρ_s orbits a planet of density ρ_p and radius R_p. (a) Show that a loose body is disrupted at a distance of order R_p(2ρ_p/ρ_s)^(1/3). (b) Evaluate it for a comet (density 600 {kg,m^(-3)}) near Jupiter (R_p = 7.15×10^7 {m}, ρ_p = 1330 {kg,m^(-3)}). (c) Explain why the ring systems of the giant planets lie inside this distance.

Worked solution →

[2] Problem 9 — Gravitational slingshot. A spacecraft moves at speed v along the orbit of a planet which moves at speed U in the same direction, and makes an elastic encounter with the planet. (a) Show that in the planet's frame the speed is unchanged. (b) Show that the spacecraft's final speed in the Solar System frame lies between |v-2U| and v+2U. (c) Explain the source of the extra kinetic energy and its effect on the planet.

Worked solution →

[3] Problem 10 — Hohmann transfer. A spacecraft is in a circular parking orbit of radius r_1 about a planet of mass M, and must reach a circular orbit of radius r_2 > r_1 using two tangential burns (a Hohmann transfer). (a) Show that the transfer ellipse has a = (r_1+r_2)/2, and find the speed change needed at each burn. (b) Show that the total Δ v is

$$\Delta v = \sqrt{(GM)/(r_1)}\left(\sqrt{(2r_2)/(r_1+r_2)} - 1\right) + \sqrt{(GM)/(r_2)}\left(1 - \sqrt{(2r_1)/(r_1+r_2)}\right). \tag{8}$$

(c) Evaluate the two Δ v values for a transfer from low Earth orbit (r_1 = 6.7×10^6 {m}) to geostationary orbit, find the transfer time, and explain why the first burn is the expensive one.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

HyperPhysics mechanics map

A compact concept map for revising definitions and standard results quickly.